

Tyler Stoen

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Education

California Polytechnic State University, San Luis Obispo

BS in Statistics, Cross-Disciplinary Minor in Data Science

- Expected Graduation Date: June 2025
- GPA: 3.89
- Summa cum laude

San Luis Obispo, CA

Sept 2021 - June 2025

California Polytechnic State University, San Luis Obispo

MS Statistics (In Progress)

San Luis Obispo, CA

Expected June 2026

Work Experience

Instructional Student Assistant - Statistics

California Polytechnic State University, San Luis Obispo - Statistics

- Explored and compared candidate affinity measures for detecting differential correlation across three experimental groups.
- Formulated a variance-based extension of pairwise differential correlation metrics for multi-group settings.
- Implemented procedure from existing statistical literature on multi-group correlation and p-value adjustment methods.

San Luis Obispo, CA

June 2025 - September 2025

Student Intern - Data Science

Stanford University School of Medicine - Davis Lab

- Cleaned, preprocessed and visualized mass cytometry data using R.
- Implemented and helped improve a user-contributed R package built for analyzing cytometry data.
- Trained and tested several different distance-based classification algorithms on mass cytometry data.
- Determined a new distance method classified patient cell types up to 5 percentage points more accurately than the previously used method.

Stanford, CA

June 2023 - September 2023

University Projects

Microorganism Diversity in Vineyard Soil

California Polytechnic State University, San Luis Obispo

- Collaborated with a client who provided soil data, developing a tailored analytical approach to extract meaningful insights.
- Implemented statistical models to assess correlations between soil characteristics across different soil treatments.
- Identified key contributors to microbial diversity in soil.
- Assessed microorganism functionality to identify similarities between treatments and generate actionable insights for vineyard management.
- **Soft Skills:** Data Analysis, Report Writing, Team Collaboration.

San Luis Obispo, CA

January 2025 - June 2025

Image Classification

California Polytechnic State University, San Luis Obispo

- Developed and implemented an EfficientNet model for image classification.
- Conducted an in-depth analysis to re-engineer the EfficientNet model by identifying critical features that the model utilized for classification.
- Designed and constructed a custom Convolutional Neural Network (CNN) to replicate the performance of the EfficientNet model.
- **Soft Skills:** Oral Presentation, Report writing, Teamwork.

San Luis Obispo, CA

November 2024 - December 2024

Breast Cancer Detection

California Polytechnic State University, San Luis Obispo

- Cleaned, visualised and analyzed raw data provided by UCI on breast mass measurements and features using Python.
- Trained several machine learning models to detect cancer in the breast mass data, classifying them as either benign or malignant.
- Determined the best model for future breast cancer detection.
- **Soft Skills:** Time Management, Teamwork, Report writing.

San Luis Obispo, CA

May 2024 - June 2024

Skills

Programming Python (TensorFlow, PyTorch), R, Java, JMP, Hadoop, Scala/Spark, SQL

Miscellaneous $\LaTeX$ (Overleaf/R Markdown), Quarto, Tableau, Microsoft Excel, Git.

Interests

Music Completed the Advanced Level of the Certificate of Merit test for piano with state honors. Continue to play for enjoyment.

Swimming Competed at club, high school and college levels. Current member of the Cal Poly Swim Club and previously served as Meet Coordinator (2024-2025).